Fuel cell

Renewable energy

- "Energy from a source that is not depleted when used".
- Renewable energy can be used over and over again
- It can be reused like the same states.
- Examples

  Air gives us wind energy.

  Water gives us hydral energy.

Non Renewable Energy

- Non-renewable energy is energy that cannot be used again once it is used.
- Energy something that cannot be reused once it has depleted.
- Examples

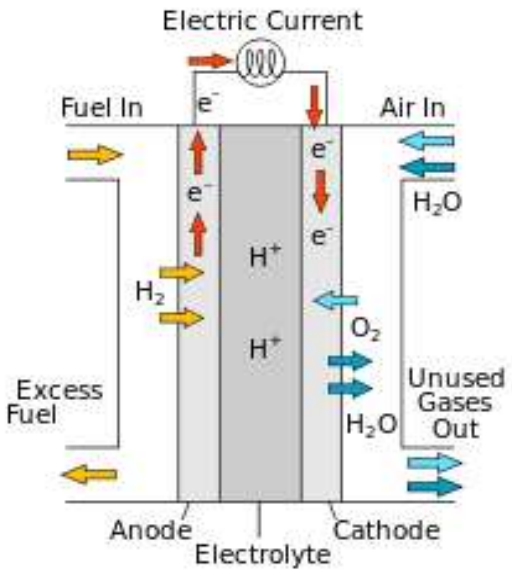
  Minerals, coals, fossil fuels

Types of renewable energy

- There are some types of renewable energies which are given below but we will mainly discuss the scope of fuel cell.
- Wind energy
- Solar energy
- Hydral energy
- Fuel cell technology

Fuel cell

- A fuel cell produces electricity through a chemical reaction, but without combustion.
- A fuel cell is an electrochemical device that combines **hydrogen** and **oxygen** to produce electricity, with water and heat as its by-product.
- In its simplest form, a single fuel cell consists of two electrodes - an anode and a cathode - with an electrolyte between them.



Fuel cell working

- The reactions that produce electricity happen at the electrodes.
- Every fuel cell has two electrodes, one positive, called the anode, and one negative, called the cathode.
- Fuel goes to the anode side, while oxygen (or just air) goes to the cathode side.
- A chemical catalyst speeds up the reactions here.

Fuel cell applications

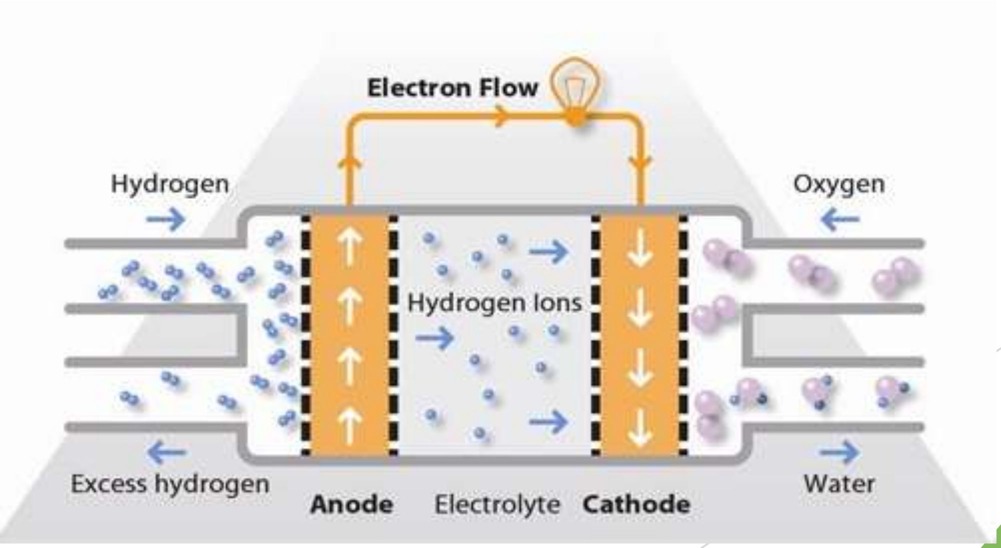
- Fuel cells are being considered for a wide range of applications.
- PEMFC have the most promising applications to buses, recreation vehicles, and lightweight vehicles.
- These are being used in portable computers.
- power systems for trains, ships and vehicles
- supplying electrical power for residential or industrial utility.
- This type of fuel cell is used in stationary power generators with output in the 100 kW to 400 kW range to power many commercial premises around the world.(PAFC)

Fuel cell types

- Proton exchange membrane fuel cell
- Solid oxide fuel cell (SOFC).
- Molten carbonate fuel cell (MCFC)
- Phosphoric acid fuel cell

Proton exchange membrane fuel cell

- “The proton exchange membrane fuel cell (PEMFC) uses a water-based, acidic polymer membrane as its electrolyte, with platinum-based electrodes”
- PEMFC cells operate at relatively low temperatures (below 100 degrees Celsius).
- Due to the relatively low temperatures and the use of precious metal-based electrodes, these cells must operate on pure hydrogen.

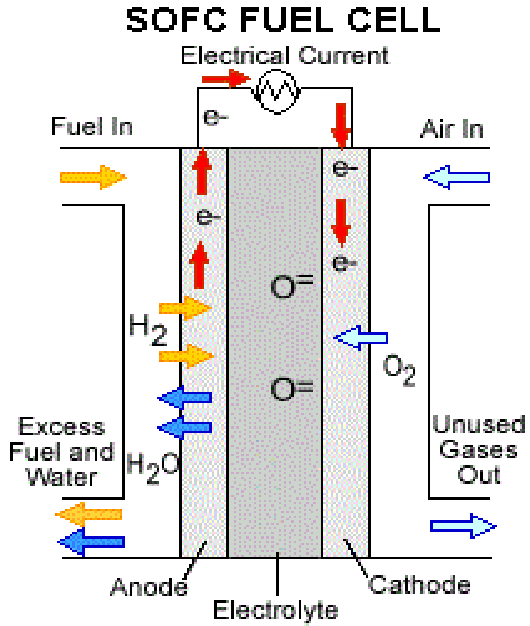


Advantages and disadvantages

- **Advantages of PEMFC**
- working temperature ~80C
- short startup time
- can work at high current densities compared to other cells.
- Disadvantages
- Low temperature
- Unefficient to use rejected heat for cogeneration of additional powers
- Difficult heat and water management limit the operating power densities

Solid oxide fuel cell

- a fuel cell that derives its energy directly from the oxidation of a solid or ceramic material called an electrolyte.
- A typical SOFC uses a solid ceramic electrolyte placed between anode (positive, and made of carbon-graphite) and cathode (negative) electrodes.
- When air, fuel, and heat are provided, the device produces electricity.



Advantages and disadvantages

Advantages

- Releasing negligible pollution.
- Since all the components are solid, as a result, there is no need for electrolyte loss maintenance and also electrode corrosion is eliminated.
- Since SOFCs are operated at high temperature, expensive catalysts such as platinum or ruthenium are totally avoided.

Disadvantages

- Fuel cell operates on high temperature, so the materials used as components are thermally challenged.
- The relatively high cost and complex fabrication are also significant problems that need to be solved.